Course Information Sheet (CIS) CHEM1043 General Chemistry II (2025 – 26 Semester 1)

(I)	**Course Coordinator:**	Dr. A T L Li (litla@hku.hk)
(II)	**Teachers:**	Prof. Seungkyu LEE (Topic 1-3, skchem@hku.hk) Dr. A T L Li (Topic 4-6, litla@hku.hk)
(III)	**Teaching Assistants: (Tutors)**	Mr. Zheqi LU (zheqilu_chem@connect.hku.hk) Ms. Chenxi XIONG (xiongchx@connect.hku.hk) Mr. Mengtian ZHANG (skydreamerzmt@connect.hku.hk)

(IV) Course Aim:

This course is a continuation of *CHEM1042 General Chemistry I*. It aims to further consolidate some of the important fundamentals of chemistry that underlie many topics and principles across the physical sciences. The course prepares students to pursue a major in chemistry or in other aspects that require a good foundation in chemistry.

(V) Course Contents & Topics:

1. Gases
   Simple gas laws; ideal gas equation; gases in chemical reactions; mixture of gases; kinetic-molecular theory of gases; diffusion and effusion; non-ideal gases.
2. Structure and Bonding: The Delocalized Approach: Molecular Orbital Theory
   Bonding in homonuclear and heteronuclear diatomic molecules of first and second period of elements; bonding in some simple polyatomic molecules; bonding in metals (band theory).
3. Solutions and their Properties
   Types of solutions; intermolecular forces and the solution process; solution formation and equilibrium; solubilities of gases; vapor pressures of solutions; osmotic pressure; freezing- point depression and boiling-point elevation of nonelectrolyte solutions; solutions of electrolytes; colloidal mixtures.
4. Solubility and Complex-Ion Equilibria
   Solubility product constant; relationship between solubility and K_{sp}; common-ion effect in solubility equilibria; limitations of the K_{sp} concept; precipitation; solubility and pH; equilibria involving complex ions; qualitative cation analysis.
5. Spontaneous Change
   Entropy change; second law and third law of thermodynamics; Gibbs free energy change of a system; Gibbs free energy of reactions; coupled reactions.
6. Electrochemistry
   Electrode potentials and their measurement; standard electrode potentials; E_{cell}, delta G, and K; E_{cell} as a function of concentrations; batteries; corrosion; electrolysis; industrial electrolysis processes.

(VI) Course Textbook:

- Petrucci; Herring; Madura; Bissonnette: General Chemistry - Principles and Modern Applications, latest edition**, Pearson.

** The latest edition should be 12th edition (12E). According to the publisher, there is a delay in delivery schedule and so **they now only have the digital version** (Mastering with e-text) for the 12E for us. Our notes will still be based on the 11E and so it is perfectly okay for students to use 11E (or 12E) to study this course.

The link to HKU Textbooks Online Ordering for purchasing the digital version for 12E is given below (for 2025-26):
https://apbookshop.com/HKU/EN/Display/PrintedBook?ISBN=9781292726175&ForMaterials=False&Created=False

(VII) Course Moodle Site & Other Learning Tools

- **CHEM1043 Course Moodle site:**

This is the **main platform** that we are using in the course. You can log in the Moodle of CHEM1043_1A_2023 through your hku portal. Course documents including this CIS document, lecture notes and other resources can be found in the Moodle. You will also submit your assignments and later collect the marked assignment through Moodle (instructions to be given to you later).

- **HKU ChemApp:**

This is a mobile app specially developed for students taking chemistry courses as an additional learning tool. Please watch the promotional video (YouTube link: https://youtu.be/kpcYguO7XZ4) to see the key features and functions of the App. You can download HKU ChemApp at

Google Play: https://play.google.com/store/apps/details?id=hk.hku.teli.ChemApp&hl=zh_HK&gl=US&pli=1

App Store: https://apps.apple.com/hk/app/hku-chemapp/id1569312776

(VIII) Course Learning Outcomes (CLOs):

On successful completion of this course, students should be able to:

CLO 1	demonstrate a knowledge and understanding of the properties and behavior of gases and apply gas laws and kinetic-molecular theory to processes involving gases
CLO 2	demonstrate a knowledge and understanding in relation to solutions and their properties, solubility and complex-ion equilibria, and also electrochemistry
CLO 3	apply molecular orbital theory to explain the formation and properties of diatomic molecules of first and second period of elements and of some simple polyatomic molecules
CLO 4	demonstrate a knowledge and understanding of the relationship between free energy and spontaneity of reaction
CLO 5	apply the theories and concepts introduced in the course to solve problems, perform calculations, make predictions and rationalize trends
CLO 6	organize and present chemical ideas in a clear, logical and coherent way
CLO 7	demonstrate awareness of the relevant applications of chemistry in society and in everyday life

(IX) Course Teaching & Learning Activities:

Lectures:	36 hours
Tutorials:	~12 hours
Reading /Self-study:	100 hours

(X) Course Assessment:

(1)	2 Assignments (30 %):	(15 %) Assignment 1 on Topics 1 – 2 ***(tentative schedule**: questions to be released by 18 Sep; completed assignment to be submitted on 9 Oct. **Check Moodle for exact arrangement**.)* (15 %) Assignment 2 on Topic 3 & part of Topic 4 ***(tentative schedule**: questions to be released by 9 Oct; completed assignment to be submitted on 6 Nov. **Check Moodle for exact arrangement**.)*
(2)	Test (20 %):	Close-book written test *(see details below ##)*
(3)	End-of-term exam (50 %):	Close-book 2-hour written exam; all topics covered in the course *(The exam will take place in Dec exam period. Note that exams are time-tabled by the central time-tabling team, not by teachers.)*

Test

Date:	11 November, 2026 *To be confirmed*
Time:	11:30 am – 12:30 pm *To be confirmed*
Scope:	Topics 1 – 3
Weighting:	20 % of the course total
Venue:	CYPP2 *To be confirmed*

(XI) Schedule for Lectures:

*(As the lectures are centrally time-tabled, students should NOT have time clashes with any other T&L activities. **All students are expected to attend the lectures in person according to the set schedule**.)*

Day	Time	Venue
Wed	11:00 am – 12:50 pm (2-hr)	CYPP2
Thu	12:00 nn – 12:50 pm (1-hr)	CYPP2

(XII) Out-of-class Learning Support & Schedule for Tutorials:

Out-of-class Learning Support

<table>
<tr><th colspan="3">Teaching Assistants for CHEM1043</th></tr>
<tr><td>Mr. Zheqi LU
(For Topics 1 – 2)
(zheqilu_chem@connect.hku.hk)</td><td>Ms. Chenxi XIONG
(For Topics 3 – 4)
(xiongchx@connect.hku.hk)</td><td>Mr. Mengtian ZHANG
(For Topics 5 – 6)
(skydreamerzmt@connect.hku.hk)</td></tr>
<tr><td></td><td>  </td><td></td></tr>
<tr><td colspan="3">Zheqi, Mengtian and Chenxi will be happy to
• answer students' questions by email and
• offer tutorials and consultation sessions to students about CHEM1043 course contents</td></tr>
</table>

If you have questions about the course contents, please seek help from Zheqi (for topics 1-2), Chenxi (for topics 3-4), and Mengtian (for topics 5-6).

There are two ways in which you can reach Zheqi/Mengtian/Chenxi:

(1) For **short questions** (that is, when they need not type a lot of words/equations in answering your enquiries), you may **send emails** to Zheqi/Mengtian/Chenxi for help. Note that at least 2 full working days should be given to them to get back to you.

(2) Through (f2f/online) **consultation sessions** where you can discuss with Zheqi/Mengtian/Chenxi directly on the questions you have got about the course. One weekly 1.5-hour consultation session (for weeks 2-6, 8-14) can be provided by Zheqi/Mengtian/Chenxi for the whole class upon request. Note that f2f consultation should be preferable to Zoom because the University has returned to f2f teaching & learning. ♦ Zheqi/Mengtian/Chenxi will arrange with students who have the request to work out the schedule. They will decide whether a particular consultation session should be f2f or online (this may depend on the number of participants and the actual situation) and they will inform students of this in advance.

♦ To allow for maximum flexibility to suit different time-tables of students, students can arrange with Zheqi/Mengtian/Chenxi a time-slot that is convenient for both sides within that specified week (Mon–Fri, except for public holidays, 9:00am–6:00pm). Students should send an email message to Zheqi/Mengtian/Chenxi at least 3 working days in advance to arrange such a consultation session. Note that first-come-first serve or the time-slot that can be made available to a larger number of students will be adopted. They have the flexibility to merge two consultation sessions (for a particular week) into one longer session to suit students' need.

See next page for the tutorial schedule

Tutorial Schedule

Note 1: **All tutorials will be recorded and the recordings be posted on Moodle by the end of the following day as far as possible**. Students are encouraged to participate in the tutorials in real time. But if some cannot do so, they may watch the recordings at a later time. [Note that given the very different time-tables of the students, there is no time-slot that can suit everyone at the same time. If you can attend the tutorial, that is good. If not, please watch the recordings and should you have questions about the contents, please feel free to contact the Tutor(s).]

Note 2: There may be changes to the following schedule to suit the actual teaching and learning pace. **You will need to regularly check Moodle/emails for any updates**. Thank you.

Date (see note 2)	Time (see note 2)	Tutor	Activity (see note 1)
15 Sep (Tue)	**4:00pm – 4:50pm** Venue: HOC412	**Mr. Zheqi LU**	**Review on Key Points in Topic 1**: Gases – Properties & Behaviour
29 Sep (Tue)	**4:00pm – 4:50pm** Venue: HOC412	**Mr. Zheqi LU**	**Review on Key Points in Topic 2**: Structure and Bonding - The Delocalized Approach: Molecular Orbital Theory
20 Oct (Tue)	**4:00pm – 5:50pm** (2-hr) Venue: HOC412	**(1st hr) Ms. Chenxi Xiong** **(2nd hr) Mr. Zheqi LU**	**Review on Key Points in Topic 3:** Solutions & Their Properties **Discussion on Assignment 1** (The marked assignment will be returned before this tutorial.)
3 Nov (Tue)	**4:00pm – 4:50pm** Venue: HOC412	**Ms. Chenxi Xiong**	**Review on Key Points in Topic 4**: Solubility and Complex-Ion Equilibria
17 Nov (Tue)	**4:00pm – 4:50pm** Venue: HOC412	**Mr. Mengtian ZHANG**	**Review on Key Points in Topic 5**: Spontaneous Change
24 Nov (Tue)	**4:00pm – 5:50pm** Venue: HOC412	**(1st hr) Ms. Chenxi XIONG** **(2nd hr) Mr. Mengtian ZHANG**	**Discussion on Test** (The marked test paper will be returned to students in this tutorial. Chenxi will also go through the answers). **Discussion on Assignment 2** (The marked assignment will be returned before this tutorial.)
28 Nov (Sat)	**Zoom** **11:00am – 12:00nn** (Zoom link to be provided and this tutorial will be recorded)	**Mr. Mengtian ZHANG**	**(Zoom)** **Review on Key Points in Topic 6**: Electrochemistry

(XIII) Study Advice

- You should attend each lecture in person.

- As far as possible, you should attend the tutorials as scheduled in person. Feel free to ask questions during or after the tutorial.

- While we will refer to the lecture notes in class to discuss the course contents, you should read also the **textbook or a suitable reference book** to consolidate your learning. **Reading textbooks/reference books is an integral part of your approach to learning in university education**.

- There are class practices (exercises) incorporated into the lecture notes for each topic covered in the course. Depending on the teaching & learning pace, we may go through some of these class practices together in class to illustrate the concepts/principles/skills taught. However, when class time does not allow, you will do these class practices at home. The answers to these practices can be found on Moodle. **For extra exercises to these class practices provided, please refer to the worked examples and also the end-of-chapter exercises of the textbook**. There is an online student solution manual for the Petrucci textbook and you can access it by using the code provided in your purchased copy to log in the book website.

- It would be good that students form **study groups** to learn from each other, to discuss issues or problems that you may have in learning the course.

- Feel free to seek help from your tutors (teaching assistants of the course) – See P.4 (XII) for details.

- Take the time to read through this document to get familiar with the course information so that you should know what you are expected to do and when to complete them (e.g., assignments, test) throughout the course.

- Develop a good habit to study your learning materials bit by bit, step by step in a timely manner. Do not procrastinate and wait until the last moment to deal with your learning! Start studying and you will often see that a task is not as difficult as you had thought. You can do it!